

1. Token & Code Embeddings

Represented Artifacts:

- Verilog / SystemC Source
- Triton / MLIR Code Tokens

Search & Inference:

- Autoregressive Sampling
- Temperature / Beam Search

Primary Latent Failure:

Vacuous / Hallucinated Syntax

2. Graph & Topo Embeddings

Represented Artifacts:

- CDFG & AIG Netlists
- Spatial Floorplan Graphs

Search & Inference:

- GNN Node/Edge Vectors
- Spatial Macro Clustering

Primary Latent Failure:

Non-Routable Congestion

3. Continuous Surrogates

Represented Artifacts:

- Discrete Microarch Tuples
- Cache / Pipeline Parameters

Search & Inference:

- Bayesian Optimization (GP)
- Acquisition (EI / UCB)

Primary Latent Failure:

Optimizer's Curse (OOD)

4. Policy & MDP States

Represented Artifacts:

- Placement Action Steps
- Tile & Loop Schedules

Search & Inference:

- Deep Q-Networks (DQN)
- MCTS & PPO Updates

Primary Latent Failure:

Reward Hacking / Exploits

INDEPENDENT EVALUATOR-OWNED PHYSICAL SIGNOFF FILTER

Candidates filtered through physical signoff engines before silicon commitment

Verilator / Yosys Linting

OpenROAD / OpenSTA P&R

SCALE-Sim / gem5 Sim

JasperGold / RedHawk-SC

VERIFIED PHYSICAL SILICON SIGNOFF & COMMITMENT

✓ DRC / LVS Clean

✓ Multi-Corner STA

✓ Dynamic IR-Drop

✓ Formal LEC Proof

✓ ATPG Fault Coverage

Human Architect & Named Commitment Authority

In-Order Retirement Control • Retains 100% Silicon Freeze & Tapeout Accountability